EPFL

NII Inter-University Research Institute Corporation Research Organization of Information and Systems National Institute of Informatics

MASTER PROJECT · M2 TARGETS

Triple Barriers & Meta-Labeling

Trade outcomes, training labels and M2 results

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READING López de Prado · AFML · Chapter 3

BASELINE Cash-MTM · M1 UP Sharpe: −2.0372

UPDATED 29 October 2026 · 48 / 48 fits reviewed

Data Splits

Split	Start	End
Train	2024-06-30	2025-08-31
Validation	2025-09-08	2025-12-31
Test	2026-01-08	2026-07-18

Test dates are entry dates. Final day-7 exits: 2026-07-25.

VERIFIED AGAINST PABLO'S PAPER

M1 Baseline

Cash-MTM · day-7 exits · 0.2% fee · valued through 2026-07-25.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−43.31%	**−2.0372**	−48.75%	1045
DOWN	12.83%	0.7622	−27.84%	2580

48 / 48
M1 baselines match the paper

28
Weekly returns
199 daily bars

−40.01%
Buy-and-hold return

UP loses money; DOWN gains. All six targets share the same M1 Test trades.

roi_ext_mtm.csv reproduced: same M1 trades, engine and final exits. Annualized seven-day Sharpe ×√(365.25/7); 28 observations.

MTM Valuation

1045 UP trades · all positions closed by 2026-07-25.

Equity calculation	Total return	Max drawdown	Sharpe
Cash · positions held at entry cost	−43.31%	−45.35%	−2.72
Cash-MTM · paper method	−43.31%	−48.75%	−2.04

Same final return

−43.31% after every trade closes.
Same trades and capital allocation.

Different path

MTM values open positions at each close.
It changes drawdown and Sharpe.

Entries end 2026-07-18 exits continue for 7 days. M1 requests 10% of free cash per trade; oversubscribed entries fill pro rata.

Pablo's Results

Kronos · 1d · TP labels · final exits 2026-07-25.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−43.31%	**−2.04**	12.83%	0.76
Random Forest	−24.66%	**−1.20**	9.74%	1.14
AutoGluon	−34.26%	**−1.53**	9.90%	1.06
CTTS	−29.60%	**−1.37**	5.83%	0.52
TabPFN	−23.87%	**−1.18**	11.67%	1.21
TabICL	−33.91%	**−1.59**	14.80%	1.36

M1 matches exactly. Our M2 models are new fits; their predictions differ from these saved models.

roi_ext_mtm.csv; 150 replay checks across 10 Kronos daily cells. Same engine, fees and exit calendar for the following target comparison.

Ground Truth

Future prices determine the label under a fixed exit rule.

AT ENTRY · FIX THE QUESTION

M1 proposes a trade
Example: UP → buy
Entry price: 100

→

Specify the trade
Profit at 104 · stop at 97
Time limit: day 7

→

Freeze the rule
Fix the exit policy
and cost assumptions.

AFTERWARDS · OBSERVE THE ANSWER

Actual market prices
100 → 101 → 97 → …
Stop hit on day 2.

→

Calculate the outcome
Buy at 100, exit at 97
−3% − 0.2% costs = −3.2%

→

Ground truth: z = 0
This proposed trade lost money.
M2 will learn to predict z.

M1 chooses the direction. The observed trade outcome determines the label.

AFML §§3.2–3.6. Illustrative long trade; assumed barrier fill; cost 0.2%.

Barriers and Meta-Labels

Define the exit

Triple-barrier method

First profit, stop or expiry.

Output: **exit time and return**.

Select the trade

Meta-labeling

Predict whether M1's proposal succeeds.

Output: **score to accept, skip or size**.

Meta-labeling works with either barrier exits or a fixed day-7 exit.

AFML §3.4 (barriers), §§3.6–3.7 (meta-labeling).

Price Path Matters



Hold until day 7

Buy at 100 → sell at 103
Net return: **+2.8%**
Meta-label: **1**

Exit at the stop

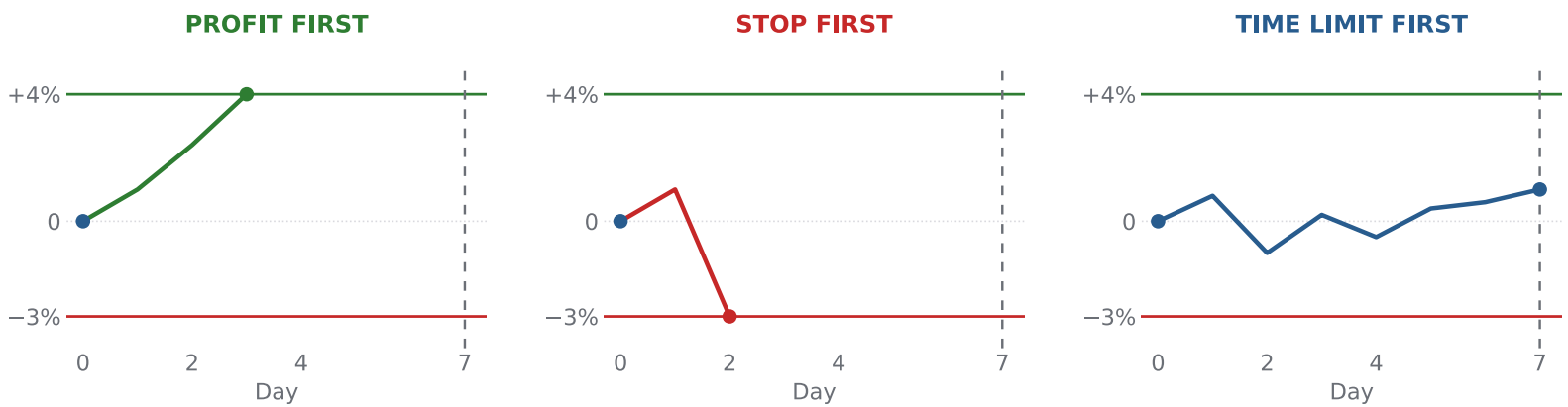
Buy at 100 → sell at 97
Net return: **−3.2%**
Meta-label: **0**

Same forecast and price path; different exit, different label.

Illustration, not market data. Profit at 104; stop at 97; time limit day 7; cost 0.2%.

First Barrier Hit

Example: buy at 100 · take profit at 104 · stop at 97 · close by day 7.



$$\tau = \min\left(t_{\text{profit}}, t_{\text{stop}}, t_{\text{expiry}}\right)$$

First touch → exit → net return → label.

AFML §3.4, pp. 45–47. Three illustrative paths, shown as gross returns on a long trade.

Setting Barrier Widths

Volatility scaling

v_0: volatility known at entry.

Profit target: $+av_0$
Stop-loss: $-bv_0$
Time limit: h bars

a and b may differ.

Our choice

$v_0 = \sigma_0\sqrt{h}, \quad h = 7$

σ_0: EWMA daily volatility from past prices.

$\sqrt{h}$ scaling is an approximation.

For a short: profit below entry, stop above.

AFML §3.3, Snippets 3.1–3.2. Experiments use a = b = k; +4%/−3% examples are illustrative.

Direction and Success

Price: **100 → 99** at expiry; no earlier barrier touch.

Market direction

Direction label: DOWN

The price fell by 1%.

Directional label: -1.

Trade success

M1 side	Net return	z
Long	−1.2%	**0**
Short	+0.8%	**1**

Falling prices can reward a short.

Label 0 = failed proposal. It does not reverse M1's direction.

AFML §§3.5–3.6; Snippets 3.5 and 3.7. Cost: 0.2%.

The Meta-Label Rule

$$g = s\left(\frac{P_\tau}{P_0} - 1\right) - c \qquad z = \begin{cases} 1 \text{ if } \ g > 0 \\ 0 \text{ otherwise} \end{cases}$$

s: +1 long / −1 short · P_0: entry · P_τ: exit · c: total cost

Long trade starting at 100	Exit price	Net return	z
Profit reached first	104	+3.8%	**1**
Stop reached first	97	−3.2%	**0**
Expiry, small gain	101	+0.8%	**1**
Expiry, loss	99	−1.2%	**0**
Expiry, gain below costs	100.1	−0.1%	**0**

Expiry can produce either label: profit = 1, otherwise = 0.

AFML Snippet 3.7; our rule subtracts 0.2% costs.

Training M2

1 · Save the inputs

At entry: market features
+ M1's forecast and side.

2 · Observe the future

Follow actual prices until
profit, stop or time limit.

3 · Calculate z

1 if net return > 0;
0 otherwise.

Repeat for historical M1 proposals

Training examples

Entry-time inputs → eventual target z
Both profitable and unprofitable proposals.

Fit M2

Learn which entry-time conditions
are associated with z = 1.

Inputs: known at entry. **Targets**: calculated from future outcomes.

AFML §§3.6–3.7. Labels can describe hypothetical trades.

Using M2

A new M1 proposal
Today's features + forecast
M1 chooses long or short.

→

Trained M2
Estimates p = P(z = 1)
for that proposed trade.

→

Decision rule
Take M1's side or skip.
Optional: determine size.

Example · take
M1: SHORT · M2: p = 0.75
With a 0.60 cutoff: take the SHORT.

Example · skip
M1: SHORT · M2: p = 0.40
With the same cutoff: no position.

M2 sees only entry-time inputs. Choose the cutoff on validation data.

AFML §§3.6–3.7 and §10.3. The 0.60 cutoff is illustrative; calibration and payoffs matter.

Trade Selection

M1 PROPOSES

100 proposed trades

55 profitable · **45** unprofitable

55%
Win rate if all proposals are taken

M2 SELECTS

40 accepted trades

32 profitable · **8** unprofitable

80%
Win rate among accepted proposals

Skipped: **23 missed winners** and **37 avoided losers**.

Higher win rate can help; profit also depends on win and loss sizes.

Illustrative counts, not results. AFML §3.7: precision–recall trade-off.

Fixed and Barrier Targets

Day-7 target

Exit: **day 7**

$$z_{\text{fixed}} = \mathbb{1}\left[s\left(\frac{P_7}{P_0} - 1\right) > c\right]$$

Did the day-7 return cover costs?

Barrier target

Exit: **first profit, stop or expiry**

$$z_{\text{TB}} = \mathbb{1}\left[s\left(\frac{P_\tau}{P_0} - 1\right) > c\right]$$

Did the barrier-exit return cover costs?

Kronos keeps the direction; the label defines what M2 learns to predict.

Current four-backend trial changes labels only; all backtests keep day-7 exits. Label rules: analysis/label_counts.py.

Return Hurdles and Exits

Target	When evaluated	What makes z = 1?
Current meta-label	Day 7	Return exceeds costs.
Larger return hurdle	Day 7	Return exceeds a volatility-scaled threshold.
Triple-barrier meta-label	First exit	Return at exit exceeds costs.

A hurdle changes the required gain. Barriers change the exit time.

AFML §§3.2–3.4. Our existing threshold experiment uses $s \cdot r_7 > k\sigma_o\sqrt{7}$; it is a separate target.

Earlier LightGBM Benchmark

LABEL COMPARISON

5.9% of labels change

Symmetric $\pm\sigma_0\sqrt{7}$ barriers.

Positive labels: **54.7% → 52.7%**

254,952 historical daily Kronos calls.

LABEL-ONLY BENCHMARK

Δ Sharpe: −0.06

Barrier target vs day-7 target; same day-7 exits.

95% interval: **−0.57 to +0.30**

This interval includes zero.

This earlier benchmark did not establish a gain.

Separate LightGBM study; daily equal-weight portfolio; 3 seeds; 20% validation cutoff. Entry dates: train 2024-07-01 – 2025-08-24 validation 2025-09-01 – 2025-12-24 Test 2026-01-01 – 2026-09-23.

Synthetic Training Labels

Real example

Observed past + actual future

Synthetic example

Generated past + generated future

→

Same label rule

Frozen M1 on past prices.
Same exits and costs on future prices.

→

Train M2

Inputs at entry
+ calculated z

Train on generated paths; confirm any benefit on held-out real data.

AFML ch. 3 applied to this thesis. Generated futures provide labels; model inputs use only the generated past.

From Outcomes to Decisions

Triple barriers
Choose the exit.

→

Ground-truth label
Did the trade profit after costs?

→

M2
Predict success for a new proposal.

Barriers: define the outcome.
Meta-labeling: select M1's proposals.

> Define the trade, calculate the labels, then test M2.

AFML §§3.4–3.7. Better prediction and profit require evidence.

READING GUIDE

Book References

Question	López de Prado (2018), AFML
Why does the price path matter?	§§3.2–3.4 · pp. 43–47
What is a directional label?	§3.5 · pp. 48–50 · Snippet 3.5
What is a meta-label?	§3.6 · pp. 50–51 · Snippet 3.7
What does the second model add?	§3.7 · pp. 51–53
How can probability inform size?	§10.3 · pp. 142–144
How do we avoid overlapping-label leakage?	§7.4 · pp. 105–109

Printed-book pages. Examples are illustrative; 0.2% costs and a 7-day horizon are our choices.

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Supporting Experiments

Earlier benchmarks and evaluation assumptions

Label Changes

Target	Positive	Negative	Labels changed
Day 7	139,453	115,499	—
Barrier k = 0.5	128,936	126,016	23.5%
Barrier k = 1	134,295	120,657	5.9%
Barrier k = 1.5	137,022	117,930	2.0%
Barrier k = 2	138,265	116,687	0.9%

Tighter barriers change more labels; performance still needs testing.

254,952 calls; all periods pooled. Entry dates: train 2024-07-01 – 2025-08-24 validation 2025-09-01 – 2025-12-24 Test 2026-01-01 – 2026-09-23. High/low touches; same-bar ties = stop. Saved label_counts.csv.

Label-Only Benchmark

M2 target	Test trades	Test Sharpe	Δ Sharpe · 95% interval
Day 7	20,976	−0.73	Reference
Barrier k = 0.5	17,386	−1.16	−0.44 · [−1.28, +0.25]
Barrier k = 1	20,398	−0.79	−0.06 · [−0.57, +0.30]
Barrier k = 1.5	20,904	−0.74	−0.01 · [−0.30, +0.19]
Barrier k = 2	20,896	−0.72	+0.00 · [−0.12, +0.10]

All intervals include zero. No confirmed improvement in this benchmark.

Separate LightGBM study; daily equal-weight portfolio; 3 seeds; 20% validation cutoff; 10-day block bootstrap. Test entries: 2026-01-01 – 2026-09-23.

Exit-Policy Benchmark

Comparison on the test period	Δ Sharpe	95% interval
M1 only: volatility barriers vs day 7	−0.67	−1.17 to −0.25
M1 only: volatility barriers vs fixed ±10%	+0.49	+0.04 to +1.06
M2: barrier label + exit vs current label + exit	−0.23	−1.32 to +0.55

Exit changes and label changes must be evaluated separately.

Separate daily equal-weight portfolio; 10-day block bootstrap; M2 uses one seed. Test entries: 2026-01-01 – 2026-09-23.

Evaluation Assumptions

Timing and validation

- Use entry-time inputs only.
- Generate M1 calls out of sample.
- Purge overlaps; apply embargoes.
- Validate choices, confirm on fresh data.

Execution assumptions

- Book snippets: closing prices.
- Our labels: daily highs and lows.
- Same-candle double touch: stop first.
- Assumed barrier fills; 0.2% costs.

AFML Snippet 3.2 and §7.4; tb.py and benchmark.py. Gaps, slippage, funding and borrow costs need further modeling.

Probability and Position Size

Prediction to position

$p = P(z = 1 \mid \text{inputs}, s)$.

A separate rule sets the size $q(p)$.

$$\text{position} = s \cdot q(p), \quad q(p) \geq 0$$

Zero size = skip; M1 keeps the side.

Payoffs matter

Win: net gain G. Loss: net cost L.

$$E[g] = pG - (1-p)L$$

$$E[g] > 0 \Leftrightarrow p > \frac{L}{G+L}$$

Gain 1, lose 2 →
break-even probability = **66.7%**.

A 50% cutoff need not break even. Account for payoffs and risk.

AFML §§3.6 and 10.3. Two-payoff example; the book uses a different sizing formula.

TP and TB Labels

Frozen M1 proposes trades; TB builds labels for M2. TP checks profit at day 7 after fees.

Label 1

Positive net return at day 7.

Label 0

Zero or negative net return at day 7.

Column	M2 training target	Purpose
TP	Original day-7 label	Baseline
TB ref	TP label on TB-eligible samples	Control for the sample filter
TB k	Net outcome at profit / stop / expiry	Test a different label

TP = true-positive meta-labeling, not take profit. All backtests keep day-7 exits.

TP rule verified on 18,230 complete proposals. Labels use future outcomes; M2 inputs use only information available at entry.

Test Sharpe

Cash-MTM · entries 2026-01-08 – 2026-07-18 · final exit 2026-07-25.
M1 Sharpe: **UP −2.0372** · DOWN 0.7622. RF = Random Forest.

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−1.329	−1.581*	−1.131	−1.638*	−1.439	−1.379
AutoGluon · DOWN	−0.126	0.695	−0.217	0.372	0.735	0.414
Kumo · UP	−1.424	−1.533	−1.666	−1.881	−1.466	−1.558
Kumo · DOWN	0.517	0.404	−1.122	0.511	0.710	0.398
TabPFN · UP	−1.651	−1.682	−1.647*	−1.834	−1.605	−1.717
TabPFN · DOWN	0.789	0.743	0.825	1.161	0.679	0.635
RF · UP	−1.105	−0.929	−1.321	−0.991	−0.839	−0.975
RF · DOWN	0.624	0.780	0.026	0.279	0.789	0.558

All UP scores are negative. Highest observed DOWN: **TabPFN k = 1 (1.161)**.

Annualized cash-MTM Sharpe; 28 weekly returns. * = original threshold fell back to 0.5. One fit per cell; exploratory results.

SAME BACKEND AND DIRECTION

Change vs TP

Δ Sharpe = target − TP · positive values mean improvement.

M2 fit	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−0.252	+0.199	−0.309	−0.110	−0.049
AutoGluon · DOWN	+0.821	−0.091	+0.498	+0.861	+0.540
Kumo · UP	−0.109	−0.243	−0.457	−0.043	−0.134
Kumo · DOWN	−0.113	−1.639	−0.006	+0.193	−0.119
TabPFN · UP	−0.031	+0.004	−0.183	+0.046	−0.067
TabPFN · DOWN	−0.047	+0.035	+0.372	−0.110	−0.154
RF · UP	+0.176	−0.216	+0.114	+0.266	+0.131
RF · DOWN	+0.155	−0.598	−0.345	+0.164	−0.066

DOWN: **AutoGluon k = 1.5 +0.861** · **TabPFN k = 1 +0.372** versus TP.

Includes the sample filter: TB removes **13 UP / 64 DOWN** early TON training proposals.

Cash-MTM through 2026-07-25 · 28 weekly returns. Same M1 trades, fees and sizing rule; fresh TP controls. Descriptive deltas.

SAME ELIGIBLE TRAINING SAMPLES

Change vs TB Reference

Δ Sharpe = TB-k − TB ref · compare labels on the same samples.

M2 fit	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	+0.451	−0.057	+0.142	+0.203
AutoGluon · DOWN	−0.912	−0.323	+0.040	−0.281
Kumo · UP	−0.134	−0.348	+0.067	−0.025
Kumo · DOWN	−1.526	+0.107	+0.306	−0.006
TabPFN · UP	+0.035	−0.152	+0.077	−0.035
TabPFN · DOWN	+0.082	+0.419	−0.064	−0.108
RF · UP	−0.392	−0.062	+0.090	−0.046
RF · DOWN	−0.753	−0.501	+0.009	−0.221

TabPFN DOWN · k = 1

Sharpe **1.161** · gain **+0.419**.
M1 DOWN: **0.762**.

RF UP · k = 1.5

Sharpe **−0.839** · gain **+0.090**.
Return **−18.15%**: still loses money.

Cash-MTM through 2026-07-25 · 28 weekly returns; one fit per cell. Gains vary by backend and direction; no optimal-k claim.